Bridge Companion: Cognitive Design System & PWA Blueprint

This document defines the intersection of **Cognitive Engineering** and **Mobile UI/UX**. It translates theoretical information organization strategies into a functional design system for a high-utility Progressive Web App (PWA).

1. Cognitive Interaction Paradigms

Applying the "Key Strategies for Information Organization" to UI Design.

A. Chunk & Prioritize (Working Memory Optimization)

- **Design Rule:** No screen should present more than 3-5 primary interactive elements or "concept blocks" at once.
- **Implementation: * Progressive Disclosure:** Use collapsible accordions for technical details.
    - **The "Core-First" Layout:** Place the "Critical Feature" (e.g., a specific Gospel bridge or response) at the top of the hierarchy, filtering out "noise" (historical context) into "Learn More" tabs.

B. Use Graphic Organizers (Visual Hierarchy)

- **Design Rule:** Use matrices and flowcharts to show relationships between Islamic beliefs and Christian responses.
- **Implementation:**
    - **The Contrast Matrix:** A simple 2-column component comparing "Islamic View" vs. "Biblical Perspective."
    - **Workflow Mapping:** Breadcrumbs and visual progress indicators that show the user exactly where they are in a conversation path.

C. Create Mental Links (Schema Integration)

- **Design Rule:** Leverage existing user knowledge of Bible structures to categorize Islamic information.
- **Implementation:**
    - **Cross-Referencing:** Active semantic links (e.g., tagging a section on "Jesus" with Biblical references) to bridge new data with familiar scripture.
    - **Consistent Iconography:** Use standard icons (cross, book, heart) to represent repeatable themes like "Salvation," "Scripture," and "Love."

D. Active Retrieval Practice (The Pressure-Test)

- **Design Rule:** Incorporate "Recall Mode" to strengthen memory under eustress.
- **Implementation:**
  - **Flashcard Toggles:** A "Quiz Me" button on specific sections that hides answers until tapped.
  - **Spaced Repetition Integration:** A "Review Later" bookmarking system that surfaces items for re-testing.

2. Visual Identity & HIG Alignment

Modern, clean, and iOS-native aesthetics to reduce extraneous cognitive load.

Typography (San Francisco / Noto Sans)

- **Headings:** Large Title (34pt), Semibold. High contrast for rapid scanning.
- **Body:** 17pt (iOS Default). Optimized line-height (1.5) to prevent "wall of text" fatigue.
- **Monospace:** Used exclusively for scripture references or specific Arabic transliterations to distinguish from body text.

Color & Emotion (Contextual Relevance)

- **Background:** `#FFFFFF` (Light) / `#1C1C1E` (Dark). Clean canvases reduce visual noise.
- **Primary Action:** `#007AFF` (iOS Blue). Familiarity leads to faster reaction times.
- **Eustress Accents:** Use `#FF9500` (Orange) for "Active Recall" or "High-Priority" notes to stimulate focus without causing panic (Red).

3. PWA Blueprint: Mobile-First Strategy

Architecture Overview

The app is a **Client-Side Only PWA** utilizing an embedded SQLite database (`sql.js`) for instant, offline-first performance.

Feature	Implementation	Cognitive Benefit
Bottom Tab Bar	4-5 core icons (Home, Topics, Search, Flashcards)	Reduces "Hick's Law" choice complexity.
Offline Mode	Service Worker + Pre-cached DB	Eliminates anxiety regarding connectivity.

Search-as-you-type	Local indexed search	Supports "Rapid Retrieval" under pressure.
Add to Home Screen	Manifest.json configuration	Reduces friction; creates a "trusted tool" mental link.

Navigation Flow (Logical Sequencing)

1. **Dashboard:** 3-5 "Quick Start" topics (e.g., "The Nature of God," "Jesus in the Quran").
2. **Category View:** Vertical list with summary "Chunks."
3. **Detail View:** Progressive disclosure using accordions.
4. **Action View:** The "Bridge" — clear steps for sharing the Gospel based on the current context.

4. Implementation Plan for Developers

Phase 1: Core Content Engine

- Initialize **Vite + React** with PWA plugin.
- Implement the **SQLite Data Layer** to parse the `BAH_info_Dump.md` into structured tables (`topics` , `scriptures` , `responses`).

Phase 2: Cognitive UI Components

- **Accordion Component:** For "Chunking" long documents.
- **Comparison Matrix:** For "Visual Mapping" of worldviews.
- **Recall Toggle:** A global state that toggles visibility of "key takeaways" for self-testing.

Phase 3: Field Optimization

- **Performance:** Ensure <1s load time via Web Workers.
- **Accessibility:** Support Dynamic Type (text resizing) and VoiceOver for users in varied lighting or audio environments.
- **Deployment:** Private GitHub repository with automated Netlify/Vercel deployment.

5. Maintenance & Retention

- **Spaced Repetition:** Log "Review Counts" in `localStorage` to surface less-frequently viewed topics.
- **Content Updates:** Simple DB swap mechanism to refresh the "info dump" without a full app rebuild.